

Predicting whole tumour genomic profiles from targeted panels with generative models

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ABSTRACT

COHORT

TASK

CODE

ESMO ID

MSK-CHORD · NSCLC

IMPACT341 → IMPACT505

github.com/chiaralouisa/missing-piece

KEY MESSAGE

A routinely available 341-gene panel carries recoverable information about genes it never assays — beyond what tumour mutational burden alone explains.

PATIENTS

2,226

NSCLC, one sample per patient

GENES PREDICTED

164

on IMPACT505, absent from IMPACT341

ALTERATION RATE

1.21%

of evaluated gene–patient pairs

BACKGROUND

Comprehensive genomic profiling is central to precision oncology, but panel gene coverage differs widely across institutions and health systems.

Smaller targeted panels cut cost and turnaround time, yet leave part of the tumour genomic profile **unmeasured** — and therefore invisible to treatment decisions.

Question. Can a conditional generative model reconstruct the missing portion of a large panel from a smaller, routinely available one?

THE COMPLETION TASK

IMPACT505 · 505 genes

IMPACT341
341 genes · OBSERVED
model input

164
PREDICTED
not assayed

The panels are nested. Everything outside the inner box is never interrogated by the small assay — that gap is what the model must fill.

COHORT

- Non-small cell lung cancer, MSK-CHORD, MSK-IMPACT sequencing
- Only patients sequenced on a panel **covering the 164 target genes** — otherwise absent calls are missing data, not wild-type
- One sample per patient, so repeat samples cannot straddle a split
- Alterations encoded as binary events: non-synonymous mutations and copy-number alterations

MODEL

Conditional flow matching

A conditional generative framework learns the joint distribution of the 164 unobserved genes given the observed 341-gene profile, rather than 164 independent classifiers.

- Continuous variant** — velocity field regressed along a straight-line path from noise to the binary profile
- Discrete variant** — flow matching directly on {0, 1, MASK}; at $t=0$ every gene is masked, so one forward pass yields exact conditional marginals with no sampling noise

Because the model is generative, it produces **coherent whole profiles** — preserving co-occurrence and mutual exclusivity — not just per-gene probabilities.

PROTOCOL

- Patient-level split, stratified on observed-panel burden: **70 / 15 / 15** **confirm split**
- AUROC as primary metric, computed **per gene** then averaged
- Bootstrap confidence intervals resampled at patient level
- Permutation test for patient-specific information

CONFIRM THE SPLIT ACTUALLY USED – THE ABSTRACT’S 75/15/15 SUMS TO 105%.

WHY THE AVERAGING MATTERS

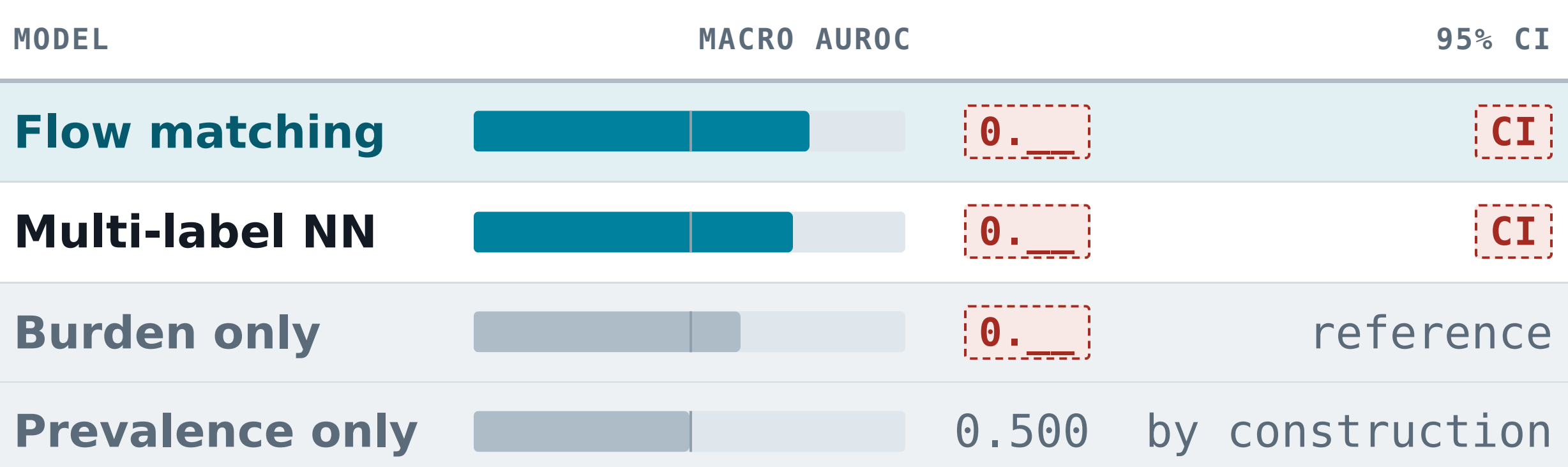
At 1.21% prevalence, **how** AUROC is averaged decides what it means.

- Pooled** — all gene–patient pairs ranked together. Dominated by prevalence differences *between* genes.
- Per-gene (macro)** — AUROC within each gene, then averaged. A patient-blind model scores exactly 0.500, so any lift is patient-specific.

We therefore report **macro AUROC**, and benchmark against a burden-only model as well as a prevalence-only baseline.

RESULTS

Performance on held-out patients



Bars show macro AUROC; the hairline marks chance (0.500). Bar widths are illustrative until the measured values are entered.

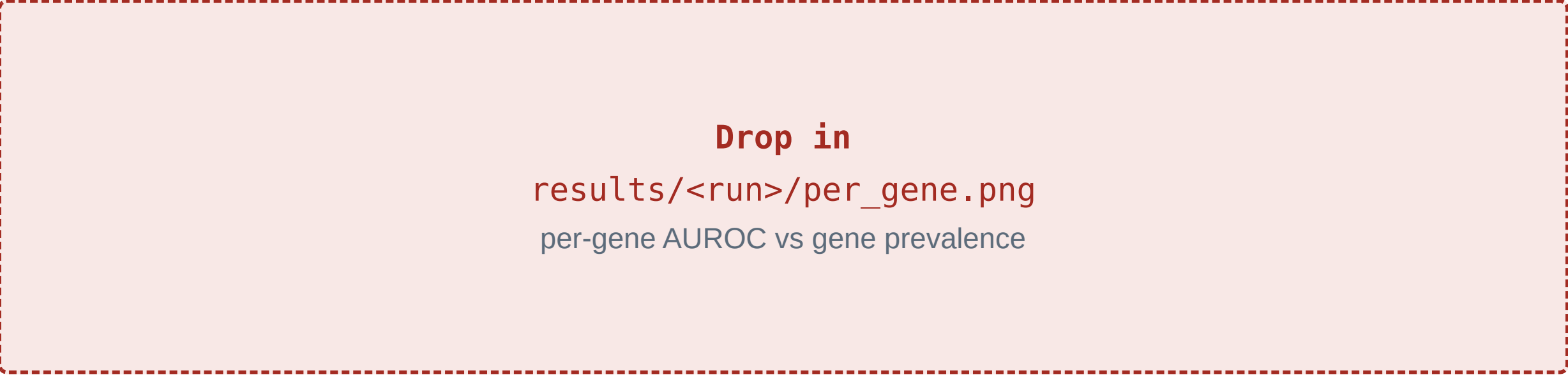
EFFECT SIZE



Performance exceeded a prevalence-only baseline in all configurations, indicating the smaller panel does carry information about alterations outside its assayed gene set.

ENTER Δ VS BURDEN – THIS IS THE NUMBER THAT SEPARATES CO-ALTERATION SIGNAL FROM MUTATIONAL BURDEN.

PER-GENE PERFORMANCE



Generated by scripts/make_figures.py. Shows whether recovery is concentrated in the commoner genes.

GENES RECOVERED

- of 164 target genes had enough positives to estimate a per-gene AUROC
- Best-recovered genes: **list top genes**
- Permutation test p **p-value** for patient-specific information

EVALUATION DESIGN

Synthetic negative controls

The pipeline was run end-to-end on simulated cohorts matched to this study’s dimensions, where the correct answer is known in advance.

REGIME	POOLED	MACRO
No signal	0.749	0.500
Burden only	0.749	0.569
True signal	0.805	0.770

On data containing **no recoverable signal at all**, pooled AUROC still reaches **0.749**, while macro AUROC correctly reads 0.500. This is why we report the per-gene average.

Simulated controls validating the evaluation, not results of this study.

LIMITATIONS

- The small-panel input is a **retrospective mask** on large-panel data, not output from a genuinely 341-gene assay
- Single-institution cohort; external validation not yet performed
- At 1.21% prevalence, positive predictive value is low at any usable sensitivity — AUROC is not a decision metric
- Predictions are **not a substitute for sequencing** and must not gate therapy

CONCLUSIONS

- Generative flow models recover clinically relevant genomic information from smaller targeted panels
- Broader tumour genomic profiles can be **partially reconstructed** from routinely available panel data
- Plausible use is **triage** — flagging tumours that warrant reflex broad sequencing — potentially broadening access to comprehensive profiling

REFERENCES

- * **Add MSK-CHORD citation (Nature 2024)**
- * **Add MSK-IMPACT citation**
- * **Add flow-matching citation (Lipman et al.)**

DISCLOSURES & FUNDING

Declare conflicts of interest and funding, or state that none were declared.
Data: MSK-CHORD, publicly available via cBioPortal under CC BY-NC.

CODE & CONTACT

QR to repo or contact

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github.com/chiaralouisa/missing-piece